Klein's proposal, therefore, space should be envisioned as having three unfurled dimensions (of which we show only two in the figure) with an additional circular dimension tacked on to each point. Notice that the extra dimension is not a bump or a loop within the usual three spatial dimensions, as the graphic limitations of the figure might lead you to think. Instead, the extra dimension is a new direction, completely distinct from the three we know about, which exists at every point in our ordinary three-dimensional space, but is so small that it escapes detection even with our most sophisticated instruments.

With this modification to Kaluza's original idea, Klein provided an answer to how the universe might have more than the three space dimensions of common experience that could remain hidden, a framework that has since become known as *Kaluza-Klein theory.* And since an extra dimension of space was all Kaluza needed to merge general relativity and electromagnetism, Kaluza-Klein theory would seem to be just what Einstein was looking for. Indeed, Einstein and many others became quite excited about unification through a new, hidden space dimension, and a vigorous effort was launched to see whether this approach would work in complete detail. But it was not long before Kaluza-Klein theory encountered its own problems. Perhaps most glaring of all, attempts to incorporate the electron into the extra-dimensional picture proved unworkable.[17] Einstein continued to dabble in the Kaluza-Klein framework until at least the early 1940s, but the initial promise of the approach failed to materialize, and interest gradually died out.

Within a few decades, though, Kaluza-Klein theory would make a spectacular comeback.

String Theory and Hidden Dimensions

In addition to the difficulties Kaluza-Klein theory encountered in trying to describe the microworld, there was another reason scientists were hesitant about the approach. Many found it both arbitrary and extravagant to postulate a hidden spatial dimension. It is not as though Kaluza was led to the idea of a new spatial dimension by a rigid chain of deductive reasoning. Instead, he pulled the idea out of a hat, and upon